Tutorial 4
STRESS STRAIN RELATIONSHIP

---

1. A nylon thread is subjected to a 8.5-N tension force. Knowing that $E = 3.3$ GPa and that the length of the thread increases by 1.1%, determine *(a)* the diameter of the thread, *(b)* the stress in the thread.

2. A block is subjected to a pressure p_o in z-direction by a rigid press as shown in Fig 1. Determine the strains and stresses, if

   (a) the deformations in x and y direction are restrained,

   (b) only the deformation in y direction is restrained,

   (c) the deformations in x and y direction are not restrained?

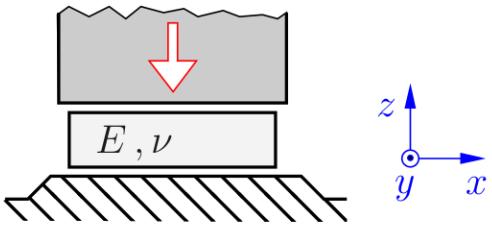


Fig. 1

3. The rubber block is confined in the U shape smooth rigid block as shown in Fig 2. If the rubber has a modulus of elasticity E and Poisson's ratio ν, determine the effective modulus of elasticity of the rubber under the confined condition.

4. An axial centric force of magnitude $P = 450\,\text{kN}$ is applied to the composite block shown by means of a rigid end plate. Knowing that $h = 10\,\text{mm}$, as shown in Fig 3. Determine the normal stress in

   (a) the brass core,

   (b) the aluminum plates.

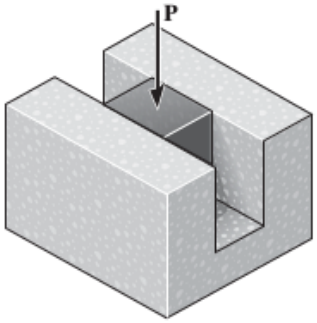


Fig. 2

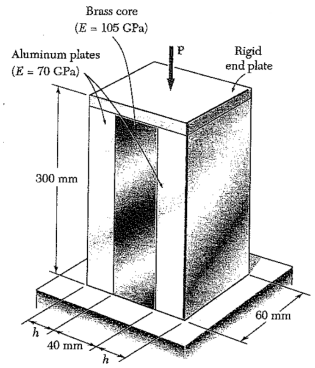


Fig. 3

5. The principal stresses at a point are shown in the Fig 4. If the material is aluminum for which and determine the principal strains. $E_{al} = 10 \times 10^3\,\text{ksi}$, $\nu_{al} = 0.33$. determine the principal strains.

6. A rectangular plate ($a > b$) is inserted into a rigid oversized opening, such that spacings of size δ are present as shown in Fig 5 . Subsequently the plate is heated. It is assumed, that the plate can move frictionless along its edges.

   a) Which temperature increase ΔT_a is required to just close the spacing on the right?

   b) For which temperature increase ΔT_b is the upper spacing just closing? What is the value of σ_x in this situation?

   c) What are the stresses in the plate for $\Delta T > \Delta T_b$?

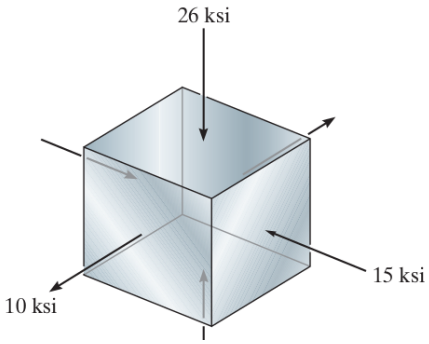


Fig. 4

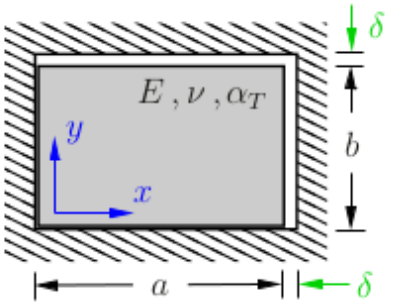


Fig. 5

7. Calculate the total elongation of the bar shown in Fig 6, under the action of self weight and end load P. Assume modulus of elasticity is E, density of material is γ, cross sectional area of the bar is A and un-deformed length is l.

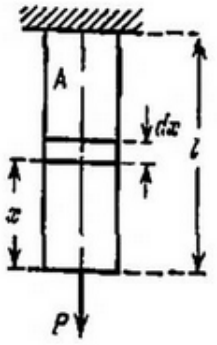


Fig. 6